PART 06 REVIEW

Evidence review: evaluate a claim, then process a system

Work from the supplied information. State your assumptions before revealing a solution.

Solution edition: includes every worked solution.

Convert a performance claim into checkable arithmetic

A fictional locked study has 10 nonoverlapping trades, each 1R $80: four gross +2R, five −1R, one −2R gap. Flat costs $4/trade. Account starts $10,000 with fixed-dollar sizing. Compute gross/net expectancy, net account return and flat break-even cost. Can it validate the rule?

Gross R = 8−5−2 = 1R; gross $80. Costs $40 = 0.5R, net $40 = 0.5R; gross expectancy 0.10R and net 0.05R. Net account return $40 ÷ $10,000 = 0.4% (gross 0.8%); break-even $80 ÷ 10 = $8 per round trip. It cannot validate the rule, and not merely because the data are invented. Net R per trade is +1.95 (×4), −1.05 (×5) and −2.05; the sample SD is about 1.66R and the standard error about 0.53R, so the rough ±2 SE band (about −1.00R to +1.10R) includes zero. Dropping one winner makes the mean −0.16R, and the $8 break-even is only twice the assumed $4 cost. Ten trades with no comparator or independent window cannot validate a rule.

Allocate simultaneous candidates

A simplified Atlas-20 variant has equity E = $50,000, of which $12,000 is cash; its four held positions have risen since entry, and they occupy four of the five slots. Use a $2 stop distance, a 0.5% price-risk allowance, a 10% notional cap measured at the limit price, and a cash allowance of (cash −

$2.50 entry fee) ÷ limit; round each allowance down to whole shares, and reserve shares × limit + $2.50. Eligible A: close 40.00, prior-20 high 39.00. Eligible B: close 20.40, prior-20 high 20.00. Rank by breakout margin, 100 × (close − prior-20 high) ÷ prior-20 high, rounded to 0.01 percentage point. Each limit is close × 1.01, rounded down to the cent. Which candidate gets the order, how many shares, and what cash reserve?

A's margin = 100 × (40.00 − 39.00) ÷ 39.00 = 2.5641…, which rounds to 2.56%; B's = 100 × 0.40 ÷ 20.00 = 2.00%, so A ranks first. A's limit = 40.00 × 1.01 = $40.40. Allowances: risk floor(0.5% × $50,000 ÷ $2) = floor(250 ÷ 2) = 125 shares; notional floor($5,000 ÷ $40.40) = 123; cash floor($11,997.50 ÷ $40.40) = 296. The smallest, 123, is the order, so the notional cap binds; planned 1R = 123 × $2 = $246, inside the $250 budget. Reserve 123 × $40.40 + $2.50 = $4,971.70, leaving $7,028.30 available. B stays eligible but gets no slot and is not re-offered later. Filled risk is fixed by the actual fill; there is no later substitution without a new decision.

Identify the research clock

A researcher logs six variants on 2015–19 data and locks one by a pre-written rule. Run once on held-out 2020–21 data, it averages +0.02R over 14 trades. After charting the held-out losers, she adds a filter: trade only if, within the prior 20 sessions, a completed low fell at least 1% below range support frozen before that low, and a close strictly above support followed within the breach session or the next three sessions, no later than the breakout's decision close. On 2020–21 the filtered version earns +0.31R over 8 trades. (a) Label both results. (b) How many variants has the search now tried, and under Chapter 25's independence model what is the chance that at least one no-edge variant passes a 5% test? (c) At which close is each filter event first known? (d) What data may evaluate the new version, and what must be frozen first?

(a) The locked result, +0.02R over 14 trades, stays the held-out result for v1 as recorded. The filtered +0.31R is development evidence for a new version, v2: its 8 trades are a subset of the 14, kept after their outcomes were seen, and the 2020–21 window shaped the filter, so it is not a held-out test.

(b) Seven variants. If each were an independent 5% test of a rule with no edge, $1 - 0.95^7 \approx 0.302$, about 30%; variants that share trades make the true figure different, but reporting only the survivor still overstates the evidence. (c) Each filter event is known only at the close that returns above support, not at the breach low; using it earlier is look-ahead. (d) Only a fresh window or a paper

forward study. Before it starts, freeze the filter definition and its recognition time, the comparator (v1 on the same window), the cost model and the decision criteria.

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